

q1

September 25, 2026

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[34]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
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[35]: df = pd.read_csv('smpregdata.csv', index_col = 0)
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[36]: df.head()
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[36]:
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	x1	x2	x3	x4	y
1	7.331693	9.660958	-2.981869	5.214284	-2.311878
2	16.888609	9.546231	18.440459	11.804018	0.994307
3	8.280643	6.442062	-0.225791	6.493211	-5.253876
4	3.827778	3.741506	-2.850664	-0.226420	20.056798
5	9.675971	4.385008	8.344576	5.979043	9.577914

```
[37]: from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

X = df.drop(columns = ['y'])
y = df['y']

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size = 0.2,
↳random_state = 1)
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[38]: features = X_train.columns
min_mse = float('inf')

for feature in features:
    candidate_model = LinearRegression()
    candidate_model.fit(X_train[[feature]], y_train)
    y_pred = candidate_model.predict(X_test[[feature]])
    curr_mse = mean_squared_error(y_test, y_pred)

    if curr_mse < min_mse:
        min_mse = curr_mse
        best_feature = feature
        best_one_feature_model = candidate_model
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print(f"Best feature: {best_feature}, Best MSE: {min_mse}")
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Best feature: x2, Best MSE: 84.4100477060547

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[39]: features_reduced = X_train.drop(columns = [best_feature]).columns
min_mse = float('inf')

for feature in features_reduced:
    features_to_use = [best_feature, feature]
    candidate_model = LinearRegression()
    candidate_model.fit(X_train[features_to_use], y_train)
    y_pred = candidate_model.predict(X_test[features_to_use])
    curr_mse = mean_squared_error(y_test, y_pred)

    if curr_mse < min_mse:
        min_mse = curr_mse
        best_features = features_to_use
        best_two_feature_model = candidate_model

print(f"Best feature: {best_features}, Best MSE: {min_mse}")
```

Best feature: ['x2', 'x4'], Best MSE: 55.63805895191378

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[40]: # sanity check that the models are accessible
pred_one_feature = best_one_feature_model.predict(X_test[[best_feature]])
pred_two_features = best_two_feature_model.predict(X_test[best_features])

print(np.isnan(pred_one_feature).sum())
print(np.isnan(pred_two_features).sum())
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